Atividade 2: Auditoria Forense de Software e Plano de Resgate Tecnico

Equipe: Dean Vinícius Palmeira de Meneses; Hilton Hunaldo Costa Silva Brito
Projeto: langfuse/langfuse

2026-05-31

1. Contexto e objetivo

Esta auditoria forense avalia a saude gerencial e arquitetural do repositorio langfuse/langfuse, com foco nos eixos do enunciado: Gestao (MPS.BR GPR), Anatomia do Codigo (SOLID/DRY) e Padroes GoF. Para cada item, apresentamos evidencia, diagnostico, classificacao de risco e recomendacao.

2. Eixo A: O Pulso da Gestao (MPS.BR GPR)

A1. Arqueologia de Issues

Evidencia (issue e discussoes): Issue #13933 e comentarios com analise de causa e proposta de mudanca de rota (novo endpoint by-id).

New issue

bug(sdk-python): get_dataset_runs() and related SDK methods fail with 400 when dataset name uses directory notation (slash) #13933

Open | #13952

simon376 opened 3 days ago

Describe the bug

`Langfuse.get_dataset_runs()`, `Langfuse.get_dataset_run()`, and any SDK method that embeds `dataset_name` as a URL path segment fail with `400 Bad Request` (empty body) when the dataset name uses directory notation (e.g. `regression/golden_v2`).

Langfuse promotes slash-separated names as a feature for organizing datasets into directories. However, the SDK constructs URLs like:

```
GET /api/public/datasets/regression/golden_v2/runs
```

The server interprets `golden_v2` as a sub-path rather than part of the dataset name. Percent-encoding the slash (`regression%2Fgolden_v2`) also fails with 400 — tested both via the SDK and via raw `httpx` requests directly against the API.

Expected behaviour: Dataset names using directory notation (containing `/`) should work correctly with all SDK methods that read run data.

Steps to reproduce

```python
from langfuse import Langfuse

lf = Langfuse()

# Dataset named "regression/golden_v2" exists in Langfuse
# (created via the UI using directory notation)

runs = lf.get_dataset_runs(dataset_name="regression/golden_\
# → ApiError: status_code: 400, body: (empty)

# Same failure for:
# lf.get_dataset_run(dataset_name="regression/golden_v2", ru
# lf.api.datasets.get_runs(dataset_name="regression/golden_\
# lf.api.datasets.get_run(dataset_name="regression/golden_v2

# Also fails when calling the REST API directly with httpx +
# GET /api/public/datasets/regression%2Fgolden_v2/runs  →  4
```

```python
# Note: these DO work (dataset_id as query param, not se
# lf.api.datasets.list()  →  can resolve "regression/golden_
# lf.api.dataset_run_items.list(dataset_id=resolved_id, run_
# The gap is specifically the run-listing / run-reading endp
```

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Diagnostico: A issue mostra investigacao tecnica detalhada, confirmando falha tanto no SDK (encoding de path) quanto na rota do servidor, e culmina em mudanca de escopo para adicionar endpoints by-id. Essa dinamica indica boa engenharia investigativa, mas revela dependencia entre repositorios (SDK + server) e impacto em compatibilidade de API.

Risco: Medio. A dependencia cruzada e a mudanca de escopo podem atrasar correcoes e gerar regressao se nao houver alinhamento formal.

Recomendacao: Formalizar criterios de aceitacao multi-repo (SDK + server), vincular issues entre repositorios e registrar a decisao arquitetural (ADR). Incluir teste de integracao cobrindo nomes com "/" e a nova rota by-id.

A2. Gestao de Riscos Ocultos (TODO/FIXME)

Evidencia (TODOs relevantes):
otelIngestionQueue.ts L215-L219 (log de blob storage)
webhooks.ts L49-L52 (versionamento da API de webhook)
webhooks.ts L568-L570 (templates customizados via UI)
compileChatMessages.ts L98-L105 (IDs de mensagens)

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Code | Issues 328 | Pull requests 339 | Discussions | Actions

main | langfuse / worker / src / queues / webhooks.ts

Go to file

wochinge fix(worker): truncate oversized GitHub dispatch payloads (#1375... b76b123 · 2 weeks ago

753 lines (671 loc) · 21.2 KB

```
import {
  InternalServerError,
  PromptWebhookOutboundSchema,
  WebhookDefaultHeaders,
  ActionExecutionStatus,
  LangfuseNotFoundError,
  JobConfigState,
  isSlackActionConfig,
  isWebhookAction,
  isGitHubDispatchAction,
  type AutomationDomain,
  type ActionDomainWithSecrets,
} from "@langfuse/shared";
import { decrypt, createSignatureHeader } from "@langfuse/shared/encryption";
import { Prisma, prisma } from "@langfuse/shared/src/db";
import {
  validateWebhookURL,
  whitelistFromEnv,
  fetchWithSecureRedirects,
  WEBHOOK_URL_VALIDATION_LOG_CONTEXT,
} from "@langfuse/shared/src/server";
import {
  TQueueJobTypes,
  QueueName,
  WebhookInput,
  getAutomationById,
  getActionByIdWithSecrets,
```

main | langfuse / worker / src / queues / webhooks.ts | Top

Code | Blame | Raw

```
import { Processor, Job } from "bullmq";
import { backOff } from "exponential-backoff";
import { env } from "../env";
import { SlackMessageBuilder } from "../features/slack/slackMessageBuilder";

// GitHub repository_dispatch client_payload: max 10 top-level properties and <64KB.
// https://docs.github.com/en/rest/repos/repos#create-a-repository-dispatch-event
const GITHUB_REPOSITORY_DISPATCH_MAX_PAYLOAD_BYTES = 64 * 1024;
const GITHUB_REPOSITORY_DISPATCH_TRUNCATION_MARKER =
  "[TRUNCATED: GitHub repository_dispatch payload exceeded size limit]";
const GITHUB_REPOSITORY_DISPATCH_TRUNCATED_FIELDS = [
  "prompt.prompt",
  "prompt.config",
];

// Handles both webhook and slack actions
export const webhookProcessor: Processor = async (
  job: Job<TQueueJobTypes[QueueName.WebhookQueue]>,
) => {
  try {
    return await executeWebhook(job.data.payload);
  } catch (error) {
    logger.error("Error executing WebhookJob", error);
    throw error;
  }
};

// TODO: Webhook outgoing API versioning
export const executeWebhook = async (
  input: WebhookInput,
  options?: { skipValidation?: boolean },
) => {
  const { projectId, automationId } = input;

  try {
    logger.debug(`Executing action for automation ${automationId}`);

    const automation = await getAutomationById({
      projectId,
      automationId,
    });

    if (!automation) {
      logger.warn(
        `Automation ${automationId} not found for project ${projectId}. We ack the job and will not ret
      );
      return;
    }

    // Route to appropriate handler based on action type
    if (automation.action.type === "WEBHOOK") {
      await executeWebhookAction({
        input,
        automation,
        skipValidation: options?.skipValidation,
      });
    } else if (automation.action.type === "SLACK") {
      await executeSlackAction({
        input,
        automation,
      });
    } else if (automation.action.type === "GITHUB_DISPATCH") {
      await executeGitHubDispatchAction({
        input,
        automation,
        skipValidation: options?.skipValidation,
      });
    } else {
      throw new InternalServerError(
        `Unsupported action type: ${automation.action.type}`,
      );
    }

    logger.debug(
      `Action executed successfully for action ${automation.action.id}`,
    );
  } catch (error) {
    logger.error("Error executing action", error);
    throw error;
  }
};

/**
 * Shared HTTP execution logic for webhook and GitHub dispatch actions
 * Handles: fetch with timeout, retry logic, status validation, execution tracking
 */
async function executeHttpAction({
  url,
  payload,
  headers,
  projectId,
```

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main

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  WebhookInput,
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  getActionByIdWithSecrets,
  getActionById,
  getConsecutiveAutomationFailures,
  SlackService,
  logger,
} from "@langfuse/shared/src/server";
import { Processor, Job } from "bullmq";
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 */
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  payload,
  headers,
  projectId,
```

Diagnostico: Existem TODOs em filas criticas (ingestao e webhooks) e em fluxo de LLM. Sao sinais de riscos tecnicos latentes que podem impactar confiabilidade e capacidade de auditoria (ex.: log de blob storage e versionamento de webhooks).

Risco: Medio/Alto. Os TODOs estao em areas de integracao com terceiros e pipelines de dados.

Recomendacao: Criar registro de riscos com priorizacao e dono, converter TODOs em issues com data-alvo e criterios de aceite, e definir politica de "no TODO without issue".

A3. Ritmo de Entrega e Code Review

Evidencia (cadencia): Grafico de contribuicoes e Pulse.
Evidencia (code review): comentario de follow-up indicando ajustes por feedback em PR #13955.

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- Pulse
- Contributors
- Community standards
- Commits
- Code frequency
- Dependency graph
- Network
- Forks

May 24, 2026 – May 31, 2026

Period: 1 week

Overview

109 Active pull requests

26 Active issues

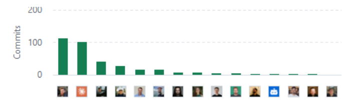
68 Merged pull requests	**41** Open pull requests	**13** Closed issues	**13** New issues

Summary

Excluding merges, **21 authors** have pushed **68 commits** to main and **237 commits** to all branches.

On main, **393 files** have changed and there have been **23,763 additions** and **9,253 deletions**

Top Committers



1 release published by **1** person

v3.176.0 v3.176.0
3 days ago

68 pull requests merged by **17** people

fix(search): match escaped unicode content in full-text search - Issue #11538
#13644 merged 2 days ago

fix: refine LANGFUSE_DISABLE_LEGACY_TRACING_IO_SEARCH handling
#13929 merged 2 days ago

perf(eval): skip FINAL and unused aggregations in checkTraceExistsAndGetTimestamp
#13934 merged 2 days ago

perf(blob-export): replace observations FINAL with LIMIT 1 BY
#13935 merged 2 days ago

feat(ui): notification for MCP v2
#13895 merged 3 days ago

fix(eval): stop retrying context overflow errors
#13930 merged 3 days ago

fix(mcp): Do not use intersection or union JSON schemas
#13927 merged 3 days ago

feat: introducing LANGFUSE_DISABLE_LEGACY_TRACING_IO_SEARCH flag to allow disabling FTS on v3 tables
#13912 merged 3 days ago

fix(mcp): Missing audit logs
#13915 merged 3 days ago

fix: tighten matches operator semantics
#13928 merged 3 days ago

fix(api,mcp): Stabilize score config pagination order
#13832 merged 3 days ago

fix(mcp): Use ids instead of names for dataset tools
#13916 merged 3 days ago

ci(deps): bump the github-actions group with 3 updates
#13924 merged 3 days ago

fix(mcp): Throw if not exists in `deleteAnnotationQueueAssignment`
#13914 merged 3 days ago

fix(mcp): Rename relevant tools from create to upsert
#13913 merged 3 days ago

feat(model-prices): add claude-opus-4-8
#13919 merged 3 days ago

fix(traces): Infinite recursion in `buildStepGroups`
#13900 merged 3 days ago

chore(mcp): Update MCP server version
#13883 merged 3 days ago

feat(ui): notification for code evals launch
#13894 merged 3 days ago

ci: notify slack on critical workflow failures
#13910 merged 3 days ago

chore(deps): dedupe and bump
#13911 merged 3 days ago

ci(sdk): use repo token for SDK spec workflow
#13909 merged 3 days ago

refactor(mcp,api): Create shared services
#13879 merged 4 days ago

fix(llm): harden LLM connection fetches
#13797 merged 4 days ago

fix(evals): tighten eval log IO cell padding
#13906 merged 4 days ago

fix(dashboards): drop scoreName filter on traces and observations views
#13901 merged 4 days ago

chore: Add .superset/ to .gitignore
#13905 merged 4 days ago

docs: Update test running instructions
#13902 merged 4 days ago

fix(evals): improve code evaluator formatting
#13897 merged 4 days ago

fix(prompts): don't comment fetching on many prompt versions

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fix(tracing): treat events-backed projects as tracing configured #13955

Open zamal-db wants to merge 2 commits into `langfuse:main` from `zamal-db:fix/hasanytrace-events-fallback`

Conversation 5 | Commits 2 | Checks 0 | Files changed 2 | +60 -16

zamal-db commented yesterday • edited

What this PR does

Improves tracing setup-state detection for events-backed ingestion.

Problem I hit

While running agent-oss style workflows on Langfuse, I hit a setup mismatch: tracing data was already present via events-backed ingestion, but setup was still showing **"tracing not configured"**.

Root cause: `hasTracingConfigured` relies on `hasAnyTrace(projectId)`, and `hasAnyTrace` was only checking ClickHouse `traces`.

How I fixed it

I updated `hasAnyTrace` to resolve tracing presence in this order:

1. PostgreSQL fast path: `project.hasTraces`
2. ClickHouse `traces`
3. ClickHouse events fallback: `hasAnyEvent(projectId)`

If either traces or events is present, it now persists `project.hasTraces = true`.

Why this matters

This removes a real false-negative setup state:

- tracing data exists,
- setup no longer says "not configured".

Type of change

- [x] Bug fix (non-breaking change which fixes an issue)
- [x] Refactor (deduplicated persistence block, no behavior change beyond fix)

Validation

Focused suite:

- `src/__tests__/server/repositories/hasAnyTrace.servertest.ts`
  - PG fast path

Reviewers

claude[bot]

+1 more reviewer

greptile-apps[bot]

Assignees

No one assigned

Labels

bug-fix-mar-26 size:S

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Diagnostico: A atividade do repositorio e continua e ha evidencias de feedback e iteracao em PRs. Entretanto, varios PRs recentes mostram apenas comentarios de bots, o que pode indicar revisao humana inconsistente.

Risco: Medio. A cadencia e boa, mas a ausencia de revisao humana explicita em parte dos PRs aumenta risco de regressao.

Recomendacao: Reforcar politica de code review (minimo 1 aprovacao humana), checklist de revisao e evidencias de testes executados em PR.

3. Eixo B: Anatomia do Codigo (SOLID/DRY)

B1. Teste da Troca (DIP)

Evidencia (pontos de acoplamento):
LLMAdapter enum
fetchLLMCompletion.ts L303-L413 (if/else por adapter)
llm-api-key router L100-L140 (branch por adapter)
useModelParams.ts L220-L269 (switch por adapter)

hassiebp feat(model-prices): add claude-opus-4-8 (#13919) ca8a928 · 3 days ago

579 lines (532 loc) · 16.6 KB

```
import { LlmApiKeys } from "@prisma/client";
import z from "zod";
import {
  BedrockConfigSchema,
  VertexAIConfigSchema,
} from "../../interfaces/customLLMProviderConfigSchemas";
import { JSONObjectSchema } from "../../utils/zod";
import type {
  InternalTraceEventInput,
  InternalTraceExperimentContext,
} from "./internalTraceEvents";

// disable lint as this is exported and used in web/worker

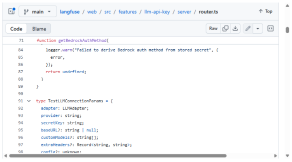
export const LLMJSONSchema = z.record(z.string(), z.any());
export type LLMJSONSchema = z.infer<typeof LLMJSONSchema>;

export const JSONSchemaFormSchema = z
  .string()
  .transform((value, ctx) => {
    try {
      const parsed = JSON.parse(value);
      return parsed;
    } catch {
      ctx.addIssue({
        code: "custom",
        message: "Parameters must be valid JSON",
      });
      return z.NEVER;
    }
  })
  .pipe(
    z
      .object({
        type: z.literal("object"),
        properties: z.record(z.string(), z.any()),
        required: z.array(z.string()).optional(),
        additionalProperties: z.boolean().optional(),
      })
      .loose()
      .transform((data) => JSON.stringify(data, null, 2)),
  );

export const LLMToolDefinitionSchema = z.object({
  name: z.string(),
  description: z.string(),
  parameters: LLMJSONSchema,
});
export type LLMToolDefinition = z.infer<typeof LLMToolDefinitionSchema>;

export const LLMToolCallSchema = z.object({
  name: z.string(),
  id: z.string(),
  args: z
    .record(z.string(), z.unknown())
    .nullable()
    .transform((val) => val ?? {}),
});
export type LLMToolCall = z.infer<typeof LLMToolCallSchema>;

export const OpenAIToolCallSchema = z.object({
  id: z.string(),
  function: z.object({
    name: z.string(),
    arguments: z.union([
      z.record(z.string(), z.unknown()),
      z
        .string()
        .transform((v) => {
          try {
            return JSON.parse(v);
          } catch {
            return v;
          }
        })
```



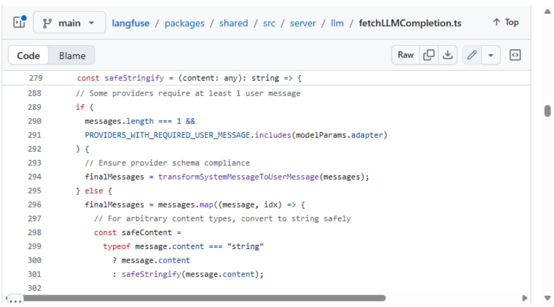
Diagnostico: A troca de provedor de IA impacta diversas camadas (shared, server e UI). A logica especifica de cada adapter esta espalhada, o que viola o Principio de Inversao de Dependencia e aumenta o custo de mudanca.

Risco: Alto. A troca de provedor exige alteracoes em varios arquivos e aumenta risco de inconsistencias.

Recomendacao: Criar um registry de provedores (Strategy/Factory), centralizando configuracao e regras por adapter; expor uma interface unica para UI consumir, reduzindo branches.

B2. Busca por "God Objects"

Evidencia (orquestracao centralizada): fetchLLMCompletion.ts L303-L413.



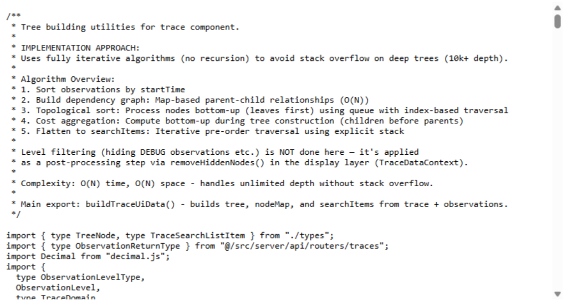
Diagnostico: O arquivo concentra configuracao de proxy, timeouts, retries e instanciacao de diferentes clientes por provider. Isso indica uma unidade com multiplas responsabilidades.

Risco: Medio/Alto. Mudancas para um provider podem afetar outros por acoplamento no mesmo fluxo.

Recomendacao: Extrair classes por provider (ex.: AnthropicStrategy, OpenAIStrategy) e deixar o orquestrador apenas selecionar e delegar.

B3. DRY (duplicacao)

Evidencia (duplicacao declarada): tree-building.ts L466 (duplicado em TraceTree e TraceTimeline).



```
/**
 * Tree building utilities for trace component.
 *
 * IMPLEMENTATION APPROACH:
 * Uses fully iterative algorithms (no recursion) to avoid stack overflow on deep trees (10k+ depth).
 *
 * Algorithm Overview:
 * 1. Sort observations by startTime
 * 2. Build dependency graph: Map-based parent-child relationships (O(N))
 * 3. Topological sort: Process nodes bottom-up (leaves first) using queue with index-based traversal
 * 4. Cost aggregation: Compute bottom-up during tree construction (children before parents)
 * 5. Flatten to searchItems: Iterative pre-order traversal using explicit stack
 *
 * Level filtering (hiding DEBUG observations etc.) is NOT done here — it's applied
 * as a post-processing step via removeHiddenNodes() in the display layer (TraceDataContext).
 *
 * Complexity: O(N) time, O(N) space - handles unlimited depth without stack overflow.
 *
 * Main export: buildTraceUiData() - builds tree, nodeMap, and searchItems from trace + observations.
 */

import { type TreeNode, type TraceSearchListItem } from "./types";
import { type ObservationReturnType } from "@/src/server/api/routers/traces";
import Decimal from "decimal.js";
import {
  type ObservationLevelType,
  ObservationLevel,
  type TraceDomain
```

Diagnostico: O proprio TODO indica duplicacao de logica. Essa repeticao tende a gerar divergencia de comportamento e custos de manutencao.

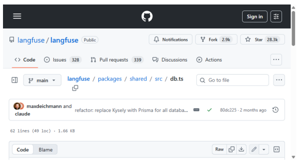
Risco: Medio.

Recomendacao: Extrair utilitario compartilhado e adicionar testes unitarios para garantir equivalencia.

4. Eixo C: Padroes de Projeto (GoF)

C1. Criacionais

Evidencia (Singleton):
PrismaClientSingleton
SlackService.getInstance



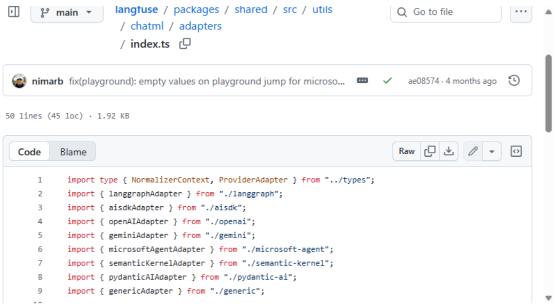
Diagnostico: O uso de Singleton garante instancia unica de recursos caros (DB e Slack). Embora seja adequado em alguns pontos, adiciona estado global e pode dificultar testes.

Risco: Baixo/Medio.

Recomendacao: Manter Singleton onde houver justificativa de custo, mas permitir injecao de dependencias em testes e documentar ciclo de vida.

C2. Estruturais

Evidencia (Adapter): chatml/adapters/index.ts L11-L37.



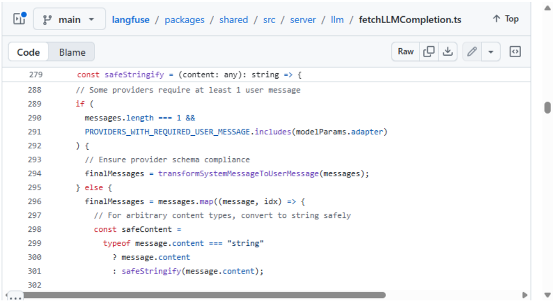
Diagnostico: O codigo organiza adapters para normalizar diferentes frameworks. Isso atende bem ao padrao Adapter, mas requer governanca (ordem, deteccao e fallback).

Risco: Baixo.

Recomendacao: Formalizar interface dos adapters, documentar criterios de deteccao e adicionar testes de regressao.

C3. Comportamentais

Evidencia (cadeia de if/else por provider): fetchLLMCompletion.ts L314-L413.



langfuse / packages / shared / src / server / llm / fetchLLMCompletion.ts

```
const safeStringify = (content: any): string => {
// Some providers require at least 1 user message
if (
  messages.length === 1 &&
  PROVIDERS_WITH_REQUIRED_USER_MESSAGE.includes(modelParams.adapter)
) {
  // Ensure provider schema compliance
  finalMessages = transformSystemMessageToUserMessage(messages);
} else {
  finalMessages = messages.map((message, idx) => {
    // For arbitrary content types, convert to string safely
    const safeContent =
      typeof message.content === "string"
        ? message.content
        : safeStringify(message.content);
```

Diagnostico: A ausencia de um Chain of Responsibility/Strategy centralizado exige alteracao no core para qualquer novo provider, o que cresce linearmente com o numero de adapters.

Risco: Medio.

Recomendacao: Aplicar Strategy + Registry (ver plano de resgate) para substituir o encadeamento de condicoes.

5. Plano de Resgate Sugerido

5.1 Refatoracao Conceitual (GoF)

Trecho alvo: `fetchLLMCompletion.ts`. O objetivo e isolar detalhes por provider e reduzir acoplamento. Exemplo conceitual:

```
interface LLMProviderStrategy {
  id: LLMAdapter
  buildClient(ctx): ChatModel
  normalizeResponse(res): NormalizedResponse
}

const registry = new Map<LLMAdapter, LLMProviderStrategy>([
  [LLMAdapter.OpenAI, new OpenAIStrategy()],
  [LLMAdapter.Anthropic, new AnthropicStrategy()],
  ...
])
```

```
export async function fetchLLMCompletion(params) {
  const strategy = registry.get(params.modelParams.adapter)
  if (!strategy) throw new Error("Unsupported adapter")
  const client = strategy.buildClient(params)
  return client.call(...)
}
```

Beneficios: reduz alteracoes no core, melhora testabilidade e permite adicionar novos providers sem editar a logica central.

5.2 Roadmap de Maturidade (MPS.BR Nivel G)

Acao 1: **GPR - Planejamento e acompanhamento:** criar baseline de planejamento (roadmap trimestral, milestones, criterios de aceite) e registrar progresso em GitHub Projects com responsaveis e metas.

Acao 2: **GRE - Rastreabilidade de requisitos:** mapear requisitos da A1 para issues/PRs, padronizar templates e adicionar links cruzados (issue -¿ PR -¿ teste).

Acao 3: **Gestao de riscos e qualidade:** manter registro de riscos (TODO/FIXME viram issues), definir politica de code review (minimo 1 aprovacao humana) e checklist de testes obrigatorios.